BIOMECHANICS OF THORAX

RIB CAGE - sternum, thoracic vertebrae, ribs
- provide stable base for muscle attachments
- protection of lungs, heart, viscera
- ventilation

KINEMATICS - ribs and manubriosternum, ribs and thoracic vertebrae.

MUSCLES FOR RESPIRATION:

ANATOMY OF STERNUM
- **MANUBRIUM** - upper bone
- **GLADIOLUS** - body
- **XIPHOID PROCESS** - lower part

STERNOCLAVICULAR JOINT - the only connection of bone and axial skeleton; which enables us to move our upper ex without disrupting thorax' anatomy.
- elevation, protraction
- essential in normal shoulder function

THORACIC CAVITY
- provide support, stable protection, and attachment.
- (image of anatomy of thorax)

RIBS (12 ribs) - helps protect the internal organs and thoracic cavity.
- (image of anatomy of thorax)
- **costal grooves** - where costal/intercostal cartilage attaches.
- **head of ribs** - located posteriorly; articulates with body of thoracic vertebrae.

MANUBRIOSTERNAL AND XIPHISTERNAL JOINTS
- external angle of louis (between t4-t5)

DIFFERENT MOTIONS:
- **pump handle motion T1-T2** - ant. and post. movement to cause expansion; movement of

COSTOTRANSVERSE JOINT (T1-T10)
(T11-T12 are both floating)

SINGLE AXIS OF THE MOTION
- believed that CV and CT joints are **mechanically interlinked.**
- single axis passing through the **centre of both joints.**
- **axis of upper ribs** - frontal plane; allows motion in sagittal plane (forward motion).
- **axis of lower ribs** - close to sagittal plane.

ANTERO-POSTERIOR MOTION *(PUMP-HANDLE MECHANISM)*

TRANSVERSE
- **back and handle motion** - 7-10th ribs
    - allows expanded motion in the lower part of the thoracic cavity (base of lungs); base —> apex.

CLINICAL IMPLICATIONS OF RIB MOTION DYSFUNCTION
- **RESTRICTED RIB MOTION (Hypomobility)**
    - postural issues, chronic respiratory conditions, muscle tightness, joint stiffness.
- **EXCESSIVE RIB MOTION (Hypermobility/Instability)**
    - ligament laxity (loose ligaments leading to proneness in fracture/dislocation); trauma; compensatory movements from spinal instability.

FIRST RIB - uppermost, smallest, and "unique"
- usual site of compression of nerve vessels (e.g. brachial plexus).
- **ANTERIORLY** - **thicker** anterior circulation.
    - **costal cartilage** - stiffer than others
    - **1st chondrosternal joint** - synchondrosis

    - **limit mobility**
    - **apex function** - where ventilation occurs (gas exchange from base —> apex).
- **POSTERIORLY** - CV joint has only **one facet.**
    - **increase mobility**

MOVEMENTS OF THE THORACIC VERTEBRAE

- depends on the orientation of **ZYGOEPIPHYSEAL** (ZP) orientation (***20 degrees*** off the frontal plane).

DIAPHRAGM - 70-80% of inspiration during **quiet breathing.**

- moves **downward**
- **circular set of muscles** arises from:
    - sternum
    - costal cartilage
    - ribs
    - vertebral bodies

INTERCOSTALS - **internal** *(exhale) and* **external intercostals** *(inhale)*

- only parasternal portions are considered as **primary muscles of ventilation.**
- subcostal group of muscles
- connects adjacent ribs to one another

ACCESSORY MUSCLES OF VENTILATION - attaches rib cage to shoulder girdle, head, vertebral column, or pelvis.

- assist with **inspiration/expiration** in situations of stress.
- When the trunk is stabilized, it moves the **vertebral column, arm,** and **head.**

NOTE:

- **PISTOL ACTION** - action of diaphragm when it contracts to allow expansion of lungs.
- **pump handle** motion, **back and handle** motion, and **pistol action** moves together (if they're not moving together, there's malfunction; e.g. *tubig sa baga*).
- **VALSALVA MANEUVER**

DIFFERENCE ASSOCIATED WITH NEONATES

- newborn has **cartilaginous**, and therefore **extremely compliant, chest wall** (develops in 3-6 months)
- **chest wall muscles** - **stabilizers**
- **rib cage** - **more horizontal alignment of ribs** (+ angle of insertion of the costal fibers)

common clinical correlation: barrel chest, pectus excavatum, carinatum